Qingjian (Anthony) Shi

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https://github.com/aqjshi/Quantitative_AI_Public

Education

University of Rochester, Hajim School of Engineering, B.S. Computer Science — May 2026

- Major GPA: 3.5/4.0.
- Courses: calculus, linear algebra, dynamic programming, blockchains, timeseries, deep learning, datamining, options pricing, discounted cash flow, financial mathematics, microeconomics, macroeconomics.

Technical Skills

Quantitative Risk & Valuation: Value-at-Risk (Historical, Parametric, Monte Carlo VaR), Expected Shortfall, Stress Testing, Scenario Analysis, Greeks (Delta, Gamma, Vega), DV01/PV01, CS01, SABR Volatility Surfaces, P&L Attribution.

Financial Systems & Data Vendors: Bloomberg Terminal, Interactive Brokers TWS, Intex (familiarity), FRED, Corporate Filings (10-K/Q via SEC EDGAR), Advanced Excel (VBA, modeling).

Languages & Scripting: Python (torch, numpy, scipy, pandas, sqlalchemy), SQL, C++, Golang, Bash/Shell Scripting, LaTeX.

Risk Infrastructure & Data Engineering: Automated ETL pipelines, asynchronous batch operations, relational schema design, Linux, Docker, AWS, telemetry dashboards, model documentation (Model Cards).

Experience

Quantitative Research Intern, RBA Engineering International – Bethesda, MD — June 2026 – Aug 2026

- Applied Kalman filters and multivariate linear regressions to bitemporal time-series to nowcast macroeconomic trends.
- Researched spurious, collinear, non-stationary time series, momentary conditional independence, causal graph theory.
- Analyzed traffic and global macroeconomic supply and transportation chains impact on traffic data.

Research Assistant, Center for Integrated Computing – Rochester, NY — Jan 2024 – May 2026

- Performed literature review and modeled quantum mechanical data, increased performance from 62% to SOTA 79%.

Projects

Macroeconomic Probabilistic Structural Premium Model, Quantitative Developer — Oct 2025 – Present

- Engineered automated balance-sheet, income statements, cash flows, and other fundamental screening pipelines across 700+ corporate issuers, extracting trailing-twelve-month (TTM) cash flows, EBITDA interest coverage, and net leverage ratios to quantify credit risk and predict structural distress regimes.
- Constructed market-neutral portfolio strategies to minimize factor drawdowns, evaluating downside tails via rolling Value-at-Risk (VaR) and Expected Shortfall metrics.
- Managed deliverables across a 3-person quant research team, orchestrating asynchronous precomputations and modular risk infrastructure to cut pipeline execution latency.
- Implemented automated daily risk reporting and performance attribution workflows, decomposing daily PnL swings into market risk sensitivities and factor exposures.
- Built robust backtesting frameworks featuring walk-forward splits, regime-shift stress testing, point-in-time validation, and out-of-sample holdouts across multi-asset universes.
- Researched and calibrated continuous time-dependent 6-parameter SABR implied volatility surfaces across multi-asset options to support derivative valuation oversight.
- Developed early-warning credit and market risk indicators to evaluate corporate distress and credit spread widening, detecting default regimes (e.g., SVB, WeWork) well in advance of structural events.
- Automated real-time risk limit monitoring and exception-alerting infrastructure using Python, Docker, and SQL, enforcing automated drawdown controls and leverage caps across 4,000+ assets.
- Designed end-to-end risk telemetry dashboards and standardized model cards to communicate quantitative assumptions, limit breaches, and stress testing findings clearly to stakeholders.